

Power for Logistic Regression

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Introduction

Power analysis for logistic regression is less standardised than for linear models. Three approaches: (1) events-per-variable rule of thumb, (2) closed-form formulas for a single predictor, (3) simulation for complex models.

Prerequisites

Logistic regression, odds ratio, events-per-variable (EPV).

Theory

EPV rule (Peduzzi et al.): at least 10 events per predictor coefficient is a working minimum. Harrell's 20 EPV is safer.

Single-predictor formula (Demidenko 2007) uses $\log(OR)$ and the baseline event rate:

$$n = \frac{(z_{1-\alpha/2} + z_{1-\beta})^2}{p_0(1-p_0) \log^2(OR) \cdot \sigma_X^2 / \pi(1-\pi)},$$

with simplifications for dichotomous X .

Simulation: specify the full data-generating process, fit the intended model, compute power across many replicates. More reliable than formulas for multiple predictors.

Assumptions

- Binary outcome.
- Logit-linear predictors.
- Sufficient separation of groups.

R Implementation

```
library(WebPower)

# Single dichotomous predictor: baseline event rate p0 = 0.2, OR = 1.5
wp.logistic(n = NULL, p0 = 0.2, p1 = 0.2 * 1.5 / (1 + 0.2 * (1.5 - 1)),
  alpha = 0.05, power = 0.80,
  family = "Bernoulli", parameter = NULL, alternative = "two.sided")

# Continuous predictor: detect OR = 1.5 per 1-SD increase
wp.logistic(n = NULL, p0 = 0.2, p1 = 0.2,
  alpha = 0.05, power = 0.80,
  family = "normal",
  alternative = "two.sided",
  parameter = log(1.5))
```

```
# EPV rule: 15 predictors, min events = 10*15 = 150
# If baseline event rate is 0.20: n = 150 / 0.20 = 750
```

Output & Results

Detecting a modest OR of 1.5 typically requires hundreds of observations; continuous predictors need somewhat less n than dichotomous, given comparable effect magnitude.

Interpretation

“To detect an OR of 1.5 per unit of the standardised predictor at 80 % power and two-sided $\alpha = 0.05$, 250 participants are required (assuming 20 % baseline event rate).”

Practical Tips

- Prefer simulation over formulas when predictors are correlated or the model is complex.
- Rare-event outcomes dramatically inflate required n ; power is approximately symmetric around $p_0 = 0.5$.
- For ordinal or multinomial logistic, scale up by roughly $k - 1$ (number of contrasts).
- Penalised logistic (ridge, Firth) handles small samples with separation.
- Missing data and measurement error further increase required n .

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren't.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Sample size, power, and Quarto reporting
- Power: closed-form and simulation